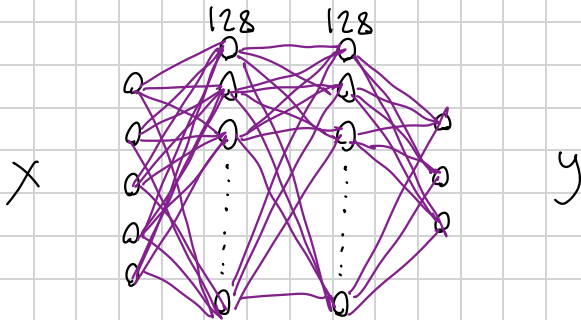


Neural Network IK 3 link



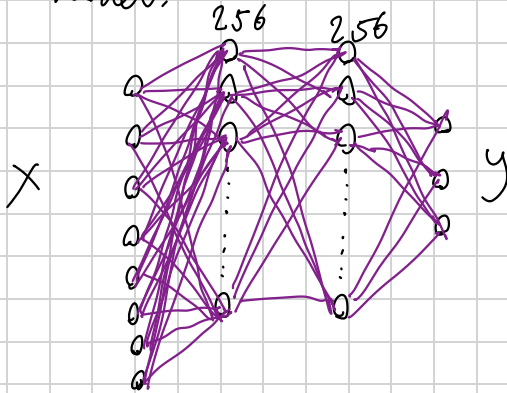
Input X: target x, target y, theta1, theta2, theta3

output Y: delta theta1, delta theta2, delta theta3

Ideas and explanation behind structure and training remains the same as 2-link, check other file for more info.

This NN model design yielded poor results. 13.2% Success, 0.442 mean error, and 19.4 ms mean time.

New model:



Still uses target x, target y, $\theta_1, \theta_2, \theta_3$, but converts into cartesian coordinates, so that we can use normalized cartesian error

X contains target x, target y, $\sin(\theta_1), \cos(\theta_1), \sin(\theta_2), \cos(\theta_2), \sin(\theta_3), \cos(\theta_3)$. 95.9% Success, 0.04 mean error, 6.16 ms